Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Augmented Robotic Manufacturing of Bio-Based Connector Kits for Modular Multi-Story Timber Buildings*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), **Advanced Robotics & Manufacturing (√),** Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical・Materials		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Production Automation Technologies*	*Robotics/Automation Machinery*	*Assembly/Precision Transfer Technologies*
Concept of Proposed Technology	Timber construction is already advanced in manufacturing and CNC adoption compared to many other building systems [1-3]. However, multi-story modular timber deployment remains constrained by (i) connection/interface performance, (ii) tolerance management across repeated production batches, and (iii) certification-grade evidence linking manufacturing controls to structural performance [4-7]. Sustainable Structural Systems Research Group at the University of Toronto is equipped with advanced expertise in robotic fabrication [8-12], bio-based construction [13, 14], computational design and full-scale experimental tests [15-18]. Integrating this expertise with Korea's design and technology development for emerging construction materials, the proposed technology develops a			

	novel form of connector kits for modular light-weight timber structures by advancing robotic manufacturing and design automation. The technology will be enabled through multi-axis fabrication, human-robot collaborative setup, optimization of geometry, finite element analysis, and structural testing for full-scale multi-story timber modular construction in South Korea. **Core concept and functions:** *1- Development of a modular, robotically-fabricated connector kits* We will develop a catalog of standardized, repeatable connector kits designed for robotic production, fast on-site assembly, and controlled tolerances. We will design pre-fabricating self-interlocking elements that enable the assembly of lightweight modular timber components. *2- Smart-factory integration:* The manufacturing workflow for the modular kit elements will be designed for manufacturing innovation. Specifically, we will integrate the design and fabrication into Korean SME's existing Building Information Modeling (BIM) workflow. We will develop software tools to be integrated with Building upon and advancing further the prototypical, project-specific solutions for such design-to-fabrication workflows, we will to create software that is easily accessible with a well-developed fabrication-aware design and planning front-end that integrates to a highly customisable, industry-ready back-end for fabrication. *3- Hybrid (experimental-numerical) simulation of multi-story systems:* Our aim is to realize a full-scale prototypes with commercial stakeholders to evaluate the robotic fabrication methods, software tools, structural performance, and construction methods in the design, manufacturing, planning and execution processes. We will validate the performance of the new structural system through a set of hybrid experimental-numerical simulation and testing, to accelerate certification-grade confidence and de-risk multi-story deployment. **Fit with the SME's demonstrated trajectory:** We look for an SME company that is capable of highly automated manufacturing systems for construction, has a prefabricated production environment where major building components are manufactured in-factory and shipped for on-site assembly, including extensive robotics and digitally controlled production with an existing/built-in Building Information Modeling (BIM) for timber construction.

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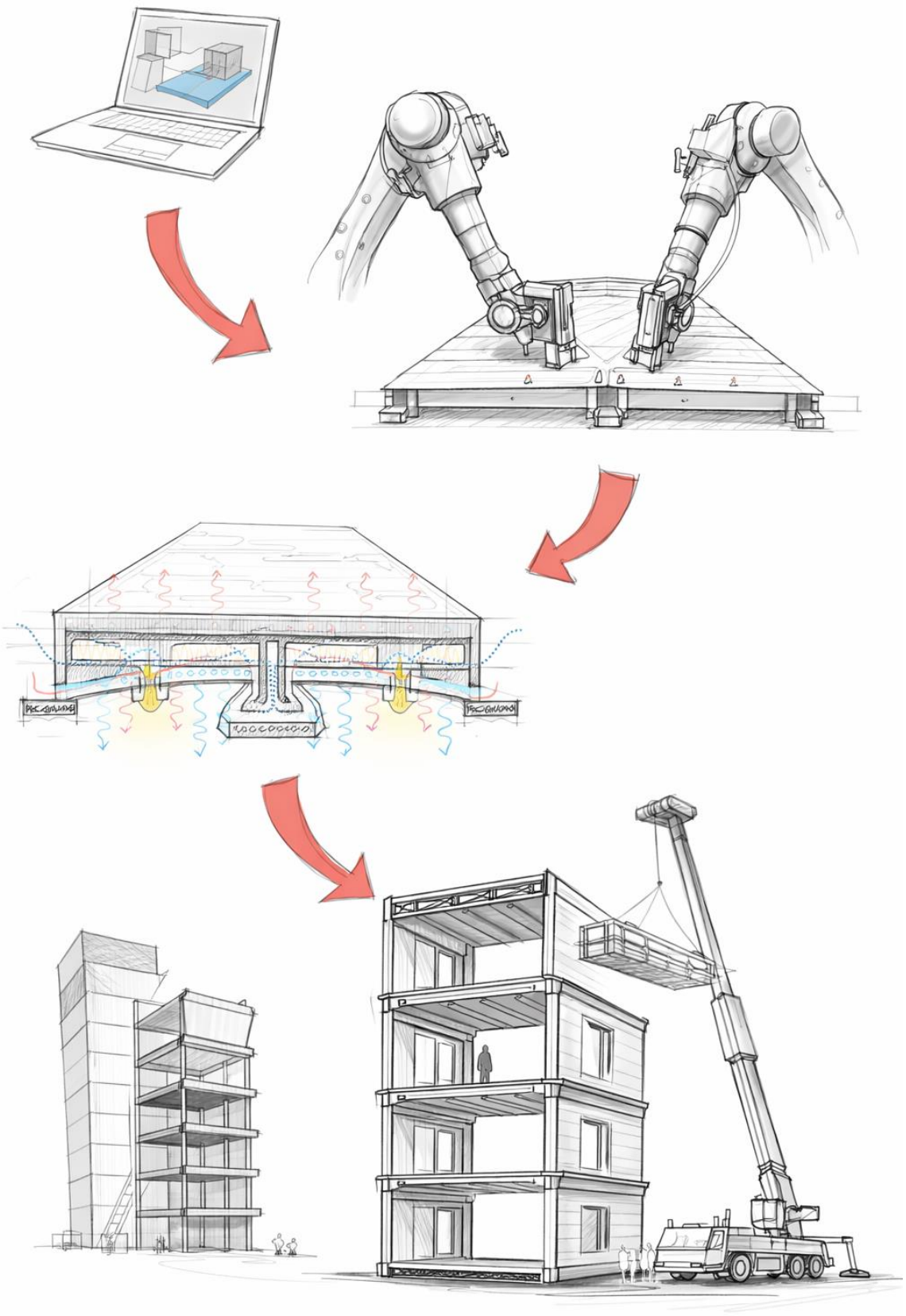

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Rationale for Joint R&D Project	A joint project is required because manufacturing scale-up and structural system validation must be developed together: • *Why collaboration is necessary:* The SME controls the production ecosystem and productization pathway (smart-factory, BIM-to-production pipeline, and modular deployment), while the research institute provides the specialized robotic fabrication methods for modular kits, hybrid simulation, experimental-computational methods, and advanced structural validation needed to responsibly scale to multi-story lightweight timber systems. This collaborative model leverages university/research infrastructure through industry–academy–research collaboration, delivering commercialization-based R&D. • *Why now:* Implementing the three core concepts/functions described above will position the SME as already operating robotic modular manufacturing with repeatability and growing ambitions (including multi-story directions), making this initiative a timely step to translate manufacturing capacity into code-ready multi-story system performance **for the first time in Korea**.
Key Words	1. Smart manufacturing innovation (robotic fabrication) 2. Digitization of manufacturing + cloud-based production/traceability 3. Process quality & productivity enhancement (in-line QC / repeatability) 4. Modular timber connector kits (Design for Manufacture and Assembly (DfMA)) 5. Hybrid simulation + experimental-computational validation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (8)
1. R&D Objectives	1. **Robotic fabrication of connector kits (algorithm and hardware development).** Develop a standardized kit-of-connectors for modular timber systems explicitly optimized for robotic fabrication, including robotic process planning, fixturing strategy, toolchain definition, and production routings that enable repeatable, high-throughput manufacturing. 2. **Verification of the workflow for fabrication of connector kits with robots.** This objective includes smart manufacturing innovation by developing a digital twin that can support product quality assessment, tolerance management, sequential assembly logic of the novel products developed in R&D Objective 1. It will implement an end-to-end digital thread for connector manufacturing and deployment, including in-line measurement, pass/fail acceptance logic, tolerance-stack management, and traceable "part passports" to drive productivity and process quality improvements. 3. **Structural performance verification of a multi-storey building employing the proposed connector kits.** The objective is to integrate the research with a first-of-its-kind adjustable multi-directional structural testing frame at University of Toronto to quantify structural performance. We will leverage robotic timber fabrication laboratory under to prototype at full scale. Together, these capabilities will allow Korean Engineering and Construction sector to deliver more housing with less waste, less on-site labour, and lower schedule risk beyond what conventional workflows can offer. 4. **Demonstration of the application of the connector kit in full-scale construction.** The objective is to scale from lab prototypes to pilot-scale prototype production within the Korean SME's smart-factory context, demonstrating stable cycle times, low defect rates, and repeatable structural performance across production batches. 5. **Development of design and manufacturing guides for the proposed connection kits** The objective is to deliver a design guide catalog, including standardized connector specifications, installation procedures, manufacturing QA/QC requirements, and verification results to support mass application of the system in Korea's timber construction industry.			
R&D Target Specifications	**A) Quantifiable Performance Targets:** 1. Advanced manufacturing innovation in construction sector by developing novel connector kits that provide interfaces for building modules. The target is to demonstrate minimal, repeatable field installation with no re-drilling/rework for building unites while allowing for multi-story construction. 2. Generate a brand-new robot fabrication "recipe" for timber construction and contribute to SME's digital model (BIM/CAD/CAM) by enabling same-day changeover from design update to robot-ready fabrication plan. 3. Demonstrate a robotic fabrication cell that can execute more than 5 connector variants with less than 10 minutes changeover (no full retooling). 4. Demonstrate a closed-loop cycle from (i) robotic prototype fabrication, (ii) physical test, (iii) digital twin update, and (iv) revised robot recipe. 5. Validate connector behavior under real-world load demands relevant to multi-story			

	modular stacks. 6. Produce a validated performance envelope for at least one representative 4-storey lightweight modular archetype where the digital twin (calibrated from component/subassembly tests) predicts global drift/residual deformation trends and identifies governing limit states at connectors. **B) Expected Intellectual Property and Scholarly Outputs (within one year after project completion)** 1. Peer-reviewed publications: at least two journal articles in SCI(E)-indexed journals. 2. Two conference papers presented in high-impact symposiums, including RobArch and World Conference in Timber Engineering. 3. Publish a manufacturing specification package defining critical dimensions, allowable tolerances, in-line QC checkpoints, acceptance criteria, and traceability requirements. 4. one patent application filed in Korea 5. Deliver a test-and-reporting package structured explicitly for prototype certification and standardization in Korea. 6. Complete at least one pilot deployment / demonstrator package for mass application.
R&D Plan	**A) Project Initiation and Standards Roadmap (Month 0–1)** a. Baseline current factory Key Performance Indicators. b. Select 3 priority modular interfaces for connector kits. **B) Connector kit architecture + digital thread ("part passport") design (Month 2–12)** a. Define connector kit-of-parts architecture and geometries. b. Define manufacturing logic, tolerances, and installation path. **C) Robotic fabrication model development + kit v1 prototype batch (Month 5–9)** a. Deliver robotic fabrication process plans ("recipes") for the interfaces. **D) Calibration of digital twin v1 (Month 10–13)** a. Complete component fabrication and sub-assembly tests for priority connectors. b. Conduct component-level destructive tests. **E) Hybrid simulation and installation validation (Month 13–21)** a. Conducting hybrid simulation scenarios to represent critical multi-story archetype behaviors (gravity + lateral + uplift interaction) and produce a system-level performance envelope. b. Validate tolerance and resilient assembly. c. Validate construction-ability. **F) Pilot-scale production run + field deployment dossier (Month 20–26)** a. Pilot-scale manufacturing run in SME environment meeting in Korea. b. Deliver a pilot demonstrator package. **G) Certification and standardization-ready package (Month 26–30)** a. Deliver the full submission-ready documentation set, b. standardized connector specifications, c. standardized installation method statement, d. qualification test and reporting protocol package (component + subassembly + physical–digital methodology).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **Technical leadership (advanced manufacturing and "Digitization of manufacturing")** a) Lead the connector kit-of-parts product architecture, including variant envelope definition (design families rather than one-off designs). b) Lead robotic fabrication design for connector concepts (geometry rationalization, assembly sequence logic, fixturing strategy, tool access, tolerance stack-up strategy), ensuring compatibility with the Korean SME's manufacturing constraints. c) Develop the assembly planning approach and define the data set required to generate robot-ready fabrication outputs from the SME's digital model. d) Specify the data structures and logic for a digital part. e) Design and execute the experimental program: component tests and sub-assembly tests required to quantify stiffness, strength, degradation, and failure modes of the robotically fabricated kits. f) Design and execute the hybrid simulation campaign to generate system-level evidence for multi-story archetypes and benchmark the digital twin from test data. g) Produce the technical content for the certification/standardization dossier, including: test protocols and reporting templates; interpretation methods; and evidence narratives linking advanced manufacturing controls to structural performance repeatability. • **Resources to be provided** a. Researchers at the master's and doctorate levels, pool of technician staff: b. Access to experimental infrastructure required for connector and subassembly and hybrid simulation testing, including instrumentation, control systems, and analysis pipelines. c. Assess to Robotic Fabrication infrastructure for timber construction
Expected R&R of Korean SME Partner	• **Manufacturing platform and commercialization leadership** a) Provide the target modular timber system definition and target use cases: module geometries, interface details, assembly sequence, current connectors/fasteners, manufacturing constraints, and site installation constraints in the Korean market. b) Lead the commercialization-oriented decisions: which kit families become product candidates, on-site construction strategy, supply chain constraints, deployment roadmap. c) Provide access to manufacturing resources for: prototype fabrication trials, iterative process tuning, and the pilot-scale production run required to demonstrate repeatability. d) Implement the robotic recipe library in the factory context, provide the operational workflow and integration support and CNC file packaging e) Lead installation trials under realistic site conditions: define crew workflow, tools, time-and-motion study plan, and field tolerance scenarios. f) Provide measured installation time per interface; document practical constraints that must be reflected in the standardized installation plan. • **Resources to be provided** a. Factory access for prototype and pilot runs; production engineering support; site installation teams; and access to BIM/CAM workflows and manufacturing data needed to enable digital thread/traceability. b. Provide technical contributions, development tasks, validation activities, resource commitments, and collaborations efforts.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver an industry-ready pathway to industrialize multi-storey modular timber construction through robot-manufactured connector kits, digitized production. The work targets SMEs operating prefabricated timber production lines with an established BIM-to-fabrication workflow, and will convert that capability into a repeatable, verifiable, and scalable manufacturing product for multi-storey deployment. **Technological achievements** The project will deliver a standardized connector kit-of-parts that transforms modular interfaces from project-specific detailing into a manufacturing-grade product family. The technology outcomes will include: • A **catalog of connector kit families** designed for robotic fabrication, fast assembly, and tolerance-managed installation across production batches. • A **digital thread / part passport framework** that links construction to robotic fabrication records and as-installed identification. • A **digital twin (physical–computational) verification workflow** that couples component and sub-assembly testing with calibrated numerical models, producing system-level performance envelopes relevant to multi-storey modular stacks. • A **standardized documentation set** (connector specifications, allowable tolerances, Quality Control checkpoints, installation guides) structured explicitly to support certification and standardization submissions. **Commercialization potential** The project is expected to generate commercialization-ready outputs that an SME can deploy immediately within an existing smart-factory and modular production environment: • The connector kit becomes a product layer with defined parametric recipes that enable repeatable manufacturing scale-up and supply-chain planning. • The verification and traceability package becomes an industry-oriented dossier that will de-risk procurement decisions and accelerating entry into regulated multi-storey applications in Korea. • By embedding manufacturability and assembly acceptance criteria into the technology, the SME can reduce rework, stabilize installation labor, and improve schedule predictability. • The project also positions the SME for licensing, strategic partnerships, and export-oriented offerings, by packaging the system as a replicable manufacturing method rather than a one-off project solution **Broader industrial and economic impact** This project is expected to strengthen national competitiveness in **advanced robotics and manufacturing for construction** by shifting multi-storey timber modular delivery from site variability to factory-controlled, data-driven production: • Establishes a replicable pathway for SMEs to industrialize multi-storey timber modular systems with robotic fabrication, along with building workforce capacity and skilled jobs. • Accelerates adoption of low-carbon construction through scalable timber modular delivery with reduced waste and improved productivity. • Enables an ecosystem effect that includes component suppliers, robot integrators, software vendors, and engineering firms can participate in a standardized modular manufacturing framework.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization